Name

Mathematics Stage 5 Core

Right-Angled Trigonometry

Book 1 Further derivatives of functions
Techniques of integration

Version: 261006

Feedback:
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Learning Intention ..... **Error! Bookmark not defined.**

Syllabus Content

MA5-TRG-C-01 applies trigonometric ratios to solve right-angled triangle problems

Demonstrate and explain the constancy of trigonometric ratios for a given angle in a right-angled triangle

- Identify and label the hypotenuse, adjacent and opposite sides with respect to a given angle in a right-angled triangle in any orientation
- Define the sine, cosine and tangent ratios for angles in right-angled triangles and use trigonometric notation $\sin\theta, \cos\theta, \tan\theta$
- Identify the sine, cosine and tangent ratios in a right-angled triangle
- Verify the constancy of the sine, cosine and tangent ratios for a given angle by applying knowledge of similar right-angled triangles
- Find approximations of the trigonometric ratios for a given angle
- Find the size of an angle given one of the trigonometric ratios for the angle using digital tools

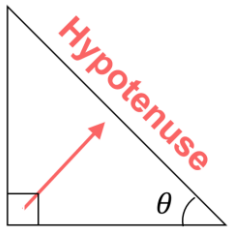
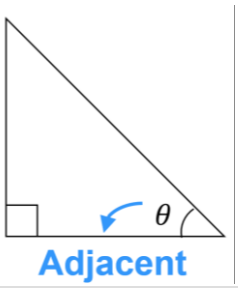
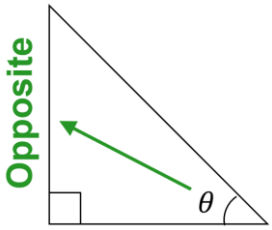
Apply trigonometry to solve right-angled triangle problems

- Apply trigonometry to find the lengths of unknown sides in right-angled triangles with a given angle including angles in degrees and minutes
- Apply trigonometry to find the size of unknown angles in right-angled triangles including in degrees and minutes
- Solve a variety of practical problems involving trigonometric ratios in right-angled triangles

Sides of a Right-angled Triangle

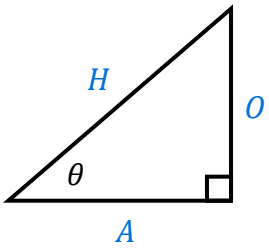
ⓘ Sides of a Right-angled Triangle

The three sides of a right-angled triangle are named based on an angle θ:

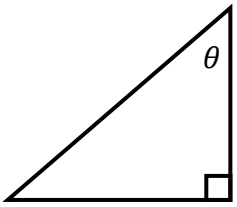
Hypotenuse	Adjacent	Opposite
Longest side Opposite the right-angle	Next to the angle, but not the hypotenuse	Furthest side from the angle
		

🔍 **Identify** hypotenuse, adjacent, and opposite sides

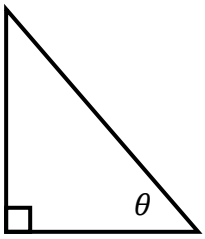
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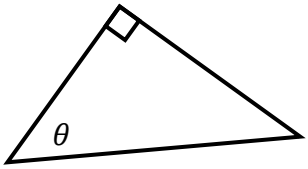
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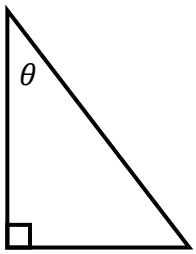
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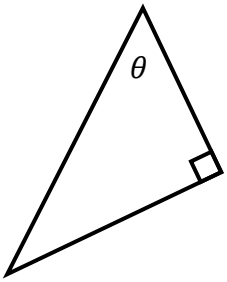
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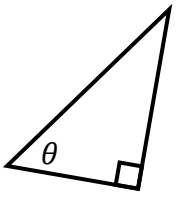
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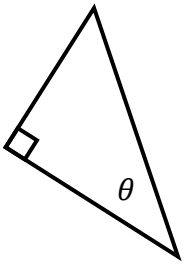
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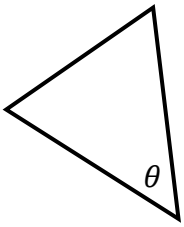
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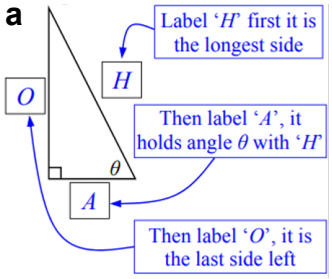
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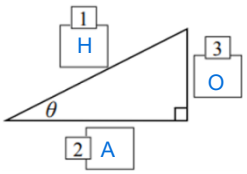
There is no H, O, A for non-right-angled triangles.

1 Label the sides of these right-angled triangles.
Remember that the hypotenuse is always opposite the right-angle.

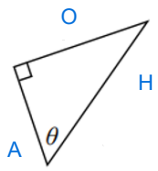
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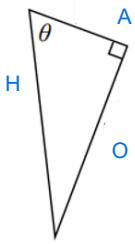
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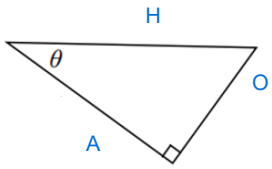
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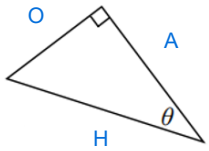
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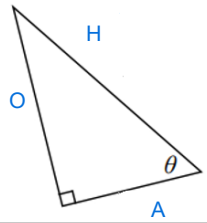
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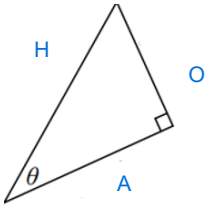
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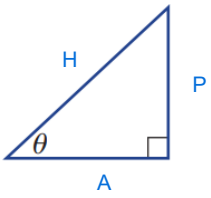
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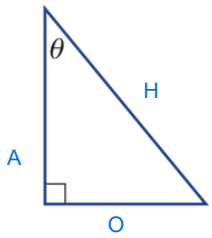
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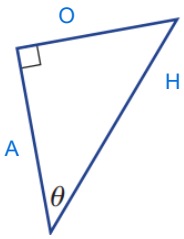
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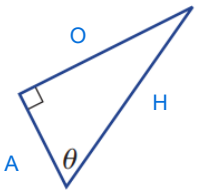
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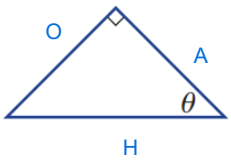
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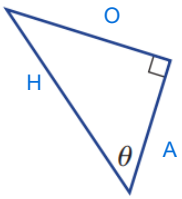
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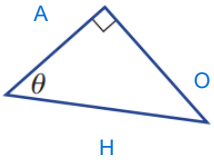
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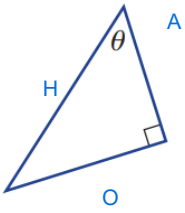
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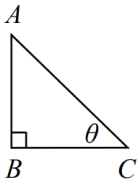


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2 Identify the hypotenuse, adjacent and opposite.

a

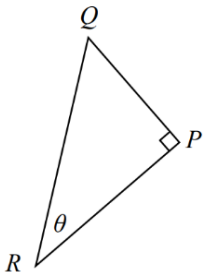


Hypotenuse: ..... AC

Adjacent: ..... BC

Opposite: ..... AB

b



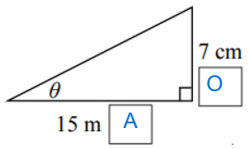
Hypotenuse: ..... QR

Adjacent: ..... RP

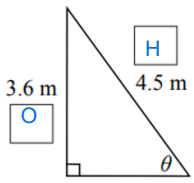
Opposite: ..... QP

3 Label the side lengths H, A, or O

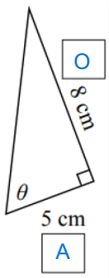
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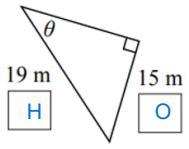
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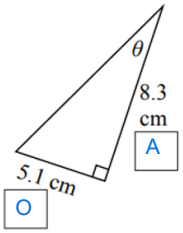
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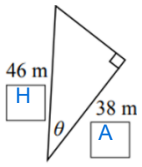
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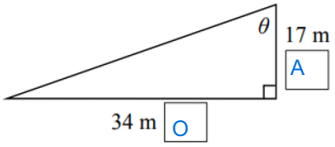
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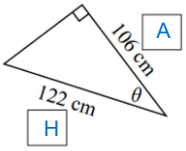
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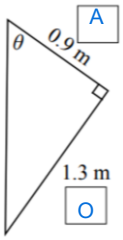
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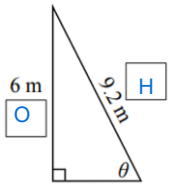
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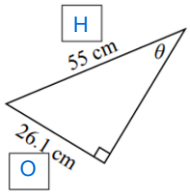
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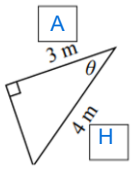
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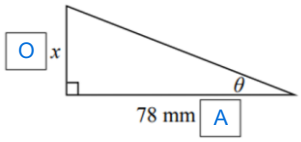
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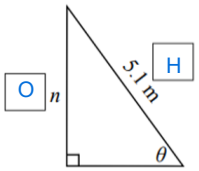
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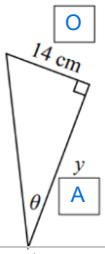
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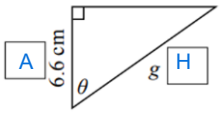
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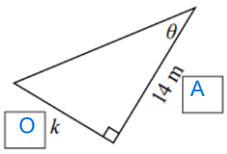
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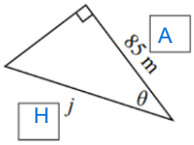
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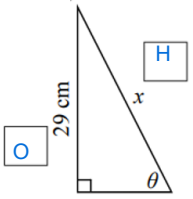
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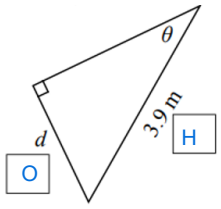
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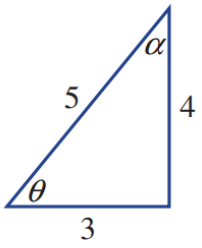
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t



4 For the triangle shown, state the length of the:



a hypotenuse 5

b side adjacent to angle θ 3

c side adjacent to angle α 4

d side opposite angle θ 4

e side opposite angle α 3

The Trigonometric Ratios

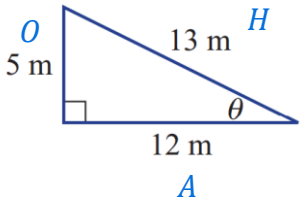
The Trigonometric Ratios

- There are three **trigonometric ratios: sine, cosine, and tangent,** which compare the lengths of two sides of a triangle.
- The ratios specify an angle, as the opposite and adjacent sides depend on the angle.
- You can remember the three ratios using SOH CAH TOA

$\sin\theta = \frac{\text{Opposite}}{\text{Hypotenuse}}$	$\cos\theta = \frac{\text{Adjacent}}{\text{Hypotenuse}}$	$\tan\theta = \frac{\text{Opposite}}{\text{Adjacent}}$

1. Identify the sides of the right-angled triangle: H, A, O
2. Write the trig ratios.

Example Identify sine, cosine, and tangent ratios



$$\sin\theta = \frac{\text{Opp}}{\text{Hyp}} = \frac{5}{13}$$

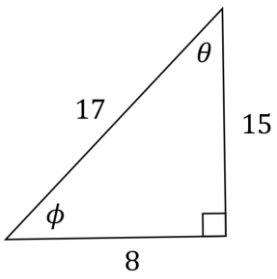
$$\cos\theta = \frac{\text{Adj}}{\text{Hyp}} = \frac{12}{13}$$

$$\tan\theta = \frac{\text{Opp}}{\text{Adj}} = \frac{5}{12}$$

Guided Practice

Identify the sine, cosine, and tangent ratios of the below triangles

a

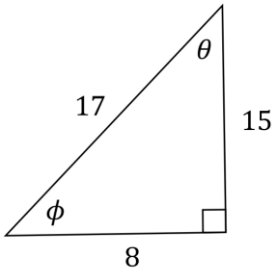


$$\sin\theta = \frac{\text{Opp}}{\text{Hyp}} = \frac{8}{17}$$

$$\cos\theta = \frac{\text{Adj}}{\text{Hyp}} = \frac{15}{17}$$

$$\tan\theta = \frac{\text{Opp}}{\text{Adj}} = \frac{8}{15}$$

b



$$\sin\phi = \frac{\text{Opp}}{\text{Hyp}} = \frac{15}{17}$$

$$\cos\phi = \frac{\text{Adj}}{\text{Hyp}} = \frac{8}{17}$$

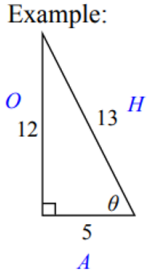
$$\tan\phi = \frac{\text{Opp}}{\text{Adj}} = \frac{15}{8}$$

Key Ideas

1. The trigonometric ratios compare lengths of two ..... of a right-angled triangle.
2. The sine ratio compares the ..... and .....
3. The cosine ratio compares the ..... and .....
4. The tangent ratio compares the ..... and .....
5. The trigonometric ratios must specify an ....., as the opposite and adjacent sides depend on the angle.

1 Label the sides, then write the ratios $\sin\theta$, $\cos\theta$, $\tan\theta$

a Example:

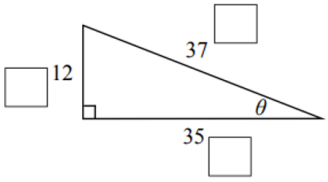


$sin\,\theta = \frac{O}{H} = \frac{12}{13}$

$cos\,\theta = \frac{A}{H} = \frac{5}{13}$

$tan\,\theta = \frac{O}{A} = \frac{12}{5}$

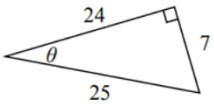
b



$\sin\theta = \frac{\square}{\square}$ $\cos\theta = \frac{\square}{\square}$ $\tan\theta = \frac{\square}{\square}$

$\sin\theta = \frac{12}{37}$, $\cos\theta = \frac{35}{37}$, $\tan\theta = \frac{12}{35}$

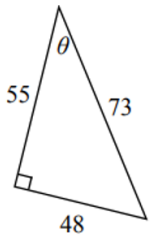
c



$\sin\theta = \frac{\square}{\square}$ $\cos\theta = \frac{\square}{\square}$ $\tan\theta = \frac{\square}{\square}$

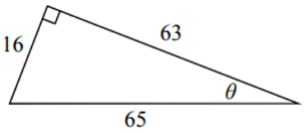
$\sin\theta = \frac{7}{25}$, $\cos\theta = \frac{24}{25}$, $\tan\theta = \frac{7}{24}$

d



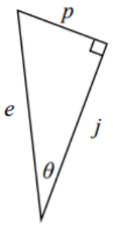
$\sin\theta = \frac{48}{73}$, $\cos\theta = \frac{55}{73}$, $\tan\theta = \frac{48}{55}$

e



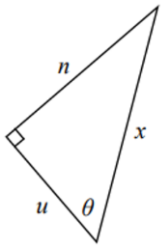
$\sin\theta = \frac{16}{65}$, $\cos\theta = \frac{63}{65}$, $\tan\theta = \frac{16}{63}$

f



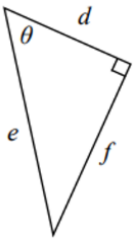
$\sin\theta = \frac{p}{e}$, $\cos\theta = \frac{j}{e}$, $\tan\theta = \frac{p}{j}$

g



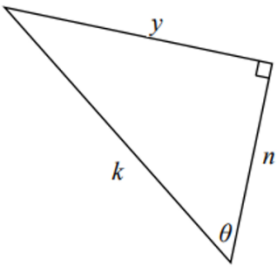
$\sin\theta = \frac{n}{x}$, $\cos\theta = \frac{u}{x}$, $\tan\theta = \frac{n}{u}$

h



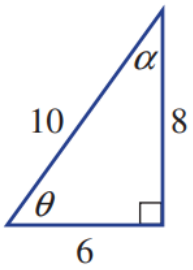
$\sin\theta = \frac{f}{e}$, $\cos\theta = \frac{d}{e}$, $\tan\theta = \frac{f}{d}$

i



$\sin\theta = \frac{y}{k}$, $\cos\theta = \frac{n}{k}$, $\tan\theta = \frac{y}{n}$

2 For the triangle shown, write the trigonometric ratio for:



a $\sin\theta =$ $\frac{8}{10} = \frac{4}{5}$

b $\cos\theta =$ $\frac{6}{10} = \frac{3}{5}$

c $\tan\theta =$ $\frac{8}{6} = \frac{4}{3}$

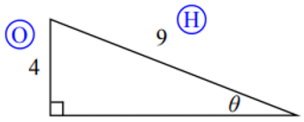
d $\sin\alpha =$ $\frac{6}{10} = \frac{3}{5}$

e $\cos\alpha =$ $\frac{8}{10} = \frac{4}{5}$

f $\tan\alpha =$ $\frac{6}{8} = \frac{3}{4}$

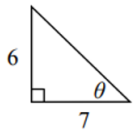
3 Label the given sides then decide the appropriate ratio that relates the two sides.

a



● sin ○ cos ○ tan

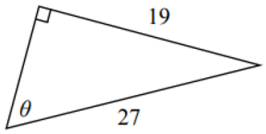
b



○ sin ○ cos ○ tan

tan

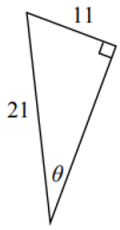
c



○ sin ○ cos ○ tan

sin

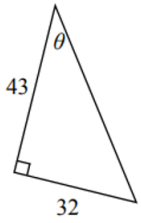
d



○ sin ○ cos ○ tan

sin

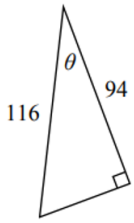
e



○ sin ○ cos ○ tan

tan

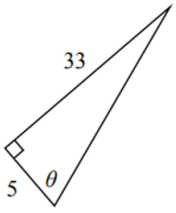
f



○ sin ○ cos ○ tan

cos

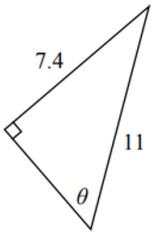
g



○ sin ○ cos ○ tan

tan

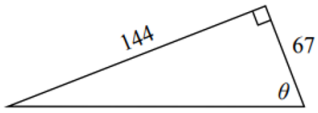
h



○ sin ○ cos ○ tan

sin

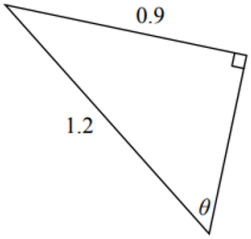
i



○ sin ○ cos ○ tan

tan

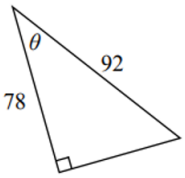
j



○ sin ○ cos ○ tan

sin

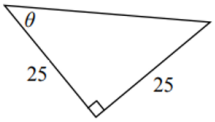
k



○ sin ○ cos ○ tan

cos

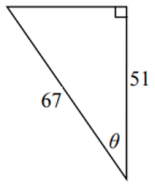
l



○ sin ○ cos ○ tan

tan

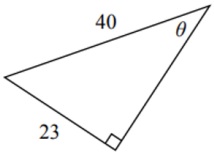
m



○ sin ○ cos ○ tan

cos

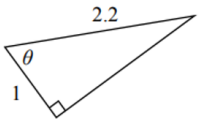
n



○ sin ○ cos ○ tan

sin

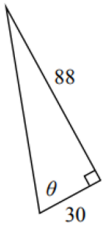
o



○ sin ○ cos ○ tan

cos

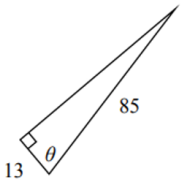
p



○ sin ○ cos ○ tan

tan

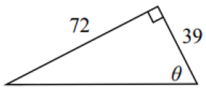
q



○ sin ○ cos ○ tan

cos

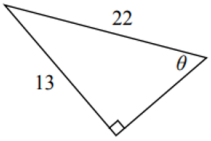
r



○ sin ○ cos ○ tan

tan

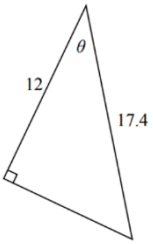
s



○ sin ○ cos ○ tan

sin

t

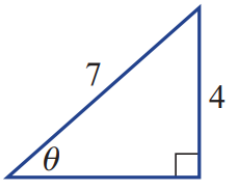


○ sin ○ cos ○ tan

cos

4 Use the given sides to write a trigonometric ratio for each triangle.

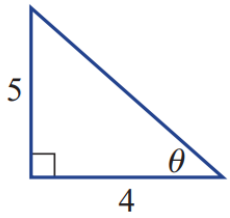
a



Given sides: O, H
Appropriate ratio: sine

$\sin\theta = \frac{4}{7}$

b

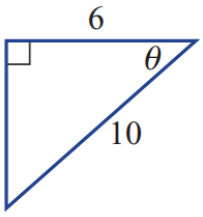


Given sides: …… , ……
Appropriate ratio: ……

…… θ = ……

$\tan\theta = \frac{5}{4}$

c

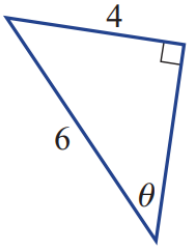


Given sides: …… , ……
Appropriate ratio: ……

……

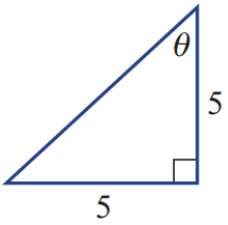
$\cos\theta = \frac{6}{10} = \frac{3}{5}$

d



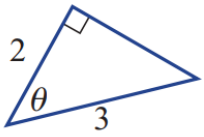
$\sin\theta = \frac{4}{6} = \frac{2}{3}$

e



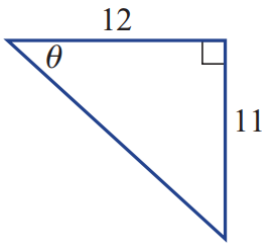
$\tan\theta = \frac{5}{5} = 1$

f



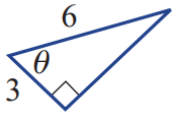
$\cos\theta = \frac{2}{3}$

g



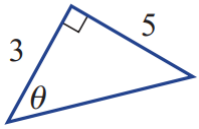
$\tan\theta = \frac{11}{12}$

h



$\cos\theta = \frac{3}{6} = \frac{1}{2}$

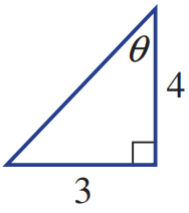
i



$\tan\theta = \frac{5}{3}$

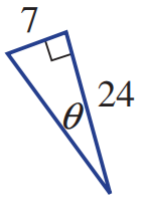
5 Use Pythagoras' Theorem to find the unknown side and then find the ratios for $\sin\theta$, $\cos\theta$, and $\tan\theta$.

a



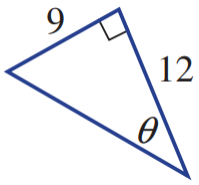
$\sin\theta = \frac{3}{5}$, $\cos\theta = \frac{4}{5}$, $\tan\theta = \frac{3}{4}$

b



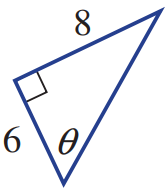
$\sin\theta = \frac{7}{25}$, $\cos\theta = \frac{24}{25}$, $\tan\theta = \frac{7}{24}$

c



$\sin\theta = \frac{9}{15} = \frac{3}{5}$, $\cos\theta = \frac{12}{15} = \frac{4}{5}$, $\tan\theta = \frac{9}{12} = \frac{3}{4}$

d



$\sin\theta = \frac{8}{10} = \frac{4}{5}$, $\cos\theta = \frac{6}{10} = \frac{3}{5}$, $\tan\theta = \frac{8}{6} = \frac{4}{3}$

Constancy of the Trigonometric Ratios

Constancy of the trigonometric ratios

Trigonometric Ratios: https://www.geogebra.org/m/k8qwer8j

For a triangle with a $30°$ interior angle, what is the ratio of:

$\sin 30° = \frac{\text{Opposite}}{\text{Hypotenuse}}$	$\cos 30° = \frac{\text{Adjacent}}{\text{Hypotenuse}}$	$\tan 30° = \frac{\text{Opposite}}{\text{Adjacent}}$

Do these values change if you make the triangle larger or smaller?

Type $\sin 30°$, $\cos 30°$, and $\tan 30°$ into your calculator.

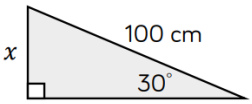
$\sin 30° =$	$\cos 30° =$	$\tan 30° =$

What do you notice?

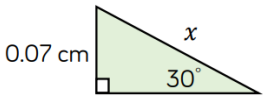
Complete the sentence:

The trigonometric ratios are always …… for a given angle, no matter the size of the triangle.

Hence, find x for these triangles:



……



……

Now we'll change the angle of the triangle. What changes, and what stays the same?

Key Ideas

1. The …… ratios remain …… for a given angle.

1 Here are two similar triangles A and B



a Explain what is meant when two figures are similar.

All angles equal.

Corresponding sides are in the same ratio.

b Write the ratio $\sin\theta$ for triangle A.

$\sin\theta = \frac{5}{13}$

c Write the ratio $\sin\theta$ for triangle B.

$\sin\theta = \frac{10}{26} = \frac{5}{13}$

d Write the ratio $\cos\theta$ for triangle A.

$\cos\theta = \frac{12}{13}$

e Write the ratio $\cos\theta$ for triangle B.

$\cos\theta = \frac{24}{26} = \frac{12}{13}$

f Write the ratio $\tan\theta$ for triangle A.

$\tan\theta = \frac{5}{12}$

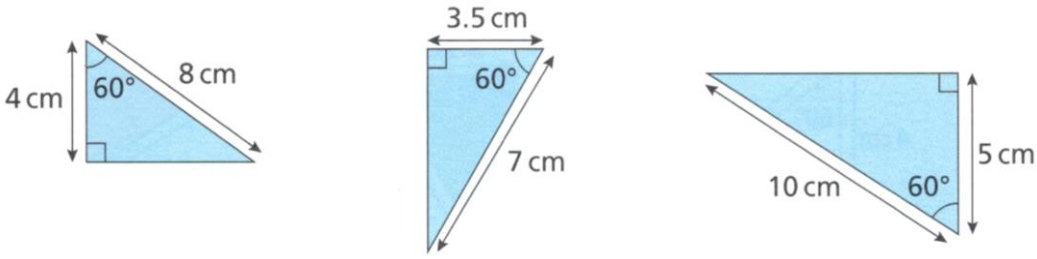
g Write the ratio $\tan\theta$ for triangle B.

$\tan\theta = \frac{10}{26} = \frac{5}{12}$

h What do you notice about your answers? (hint: fully simplify them).

The trigonometric ratios are the same for similar triangles.

2 Consider these triangles.



a Explain why these three triangles are similar.

They have the same angles: 90°, 60°, and 30° (using angle sum of a triangle)

b Calculate the ratio of $\frac{\text{Adjacent}}{\text{Hypotenuse}}$ for each of the triangles.

$\frac{4}{8} = \frac{1}{2}$, $\frac{3.5}{7} = \frac{1}{2}$, $\frac{5}{10} = \frac{1}{2}$

c What do you notice?

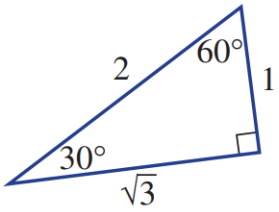
Cos ratio for all three triangles are equal.

d Type $\cos 60°$ in your calculator. What does this tell you about $\frac{\text{Adjacent}}{\text{Hypotenuse}}$ for any right-triangle with $60°$?

$\cos 60 = \frac{1}{2}$ for any right-angled triangle.

3 For the triangle shown, write the trigonometric ratio for:

a $\sin 30° =$ $\frac{1}{2}$	**b** $\cos 30° =$ $\frac{\sqrt{3}}{2}$	**c** $\tan 30° =$ $\frac{1}{\sqrt{3}}$
d $\sin 60° =$ $\frac{\sqrt{3}}{2}$	**e** $\cos 60° =$ $\frac{1}{2}$	**f** $\tan 60° =$ $\sqrt{3}$



What do you notice about cos 30° and sin 60°?

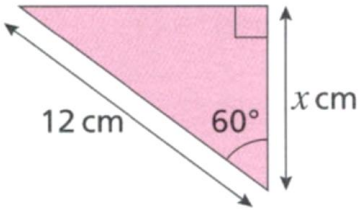
They're equal.

What do you notice about cos 60° and sin 30°?

They're equal.

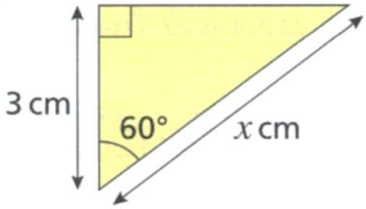
4 Use the fact that $\cos 60° = \frac{1}{2}$ to find the value of x in each of these triangles.

a



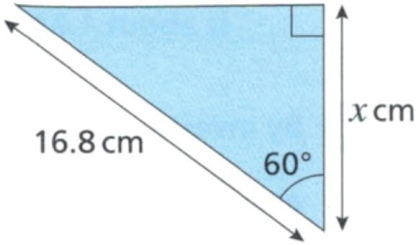
$x = 6$ cm

b



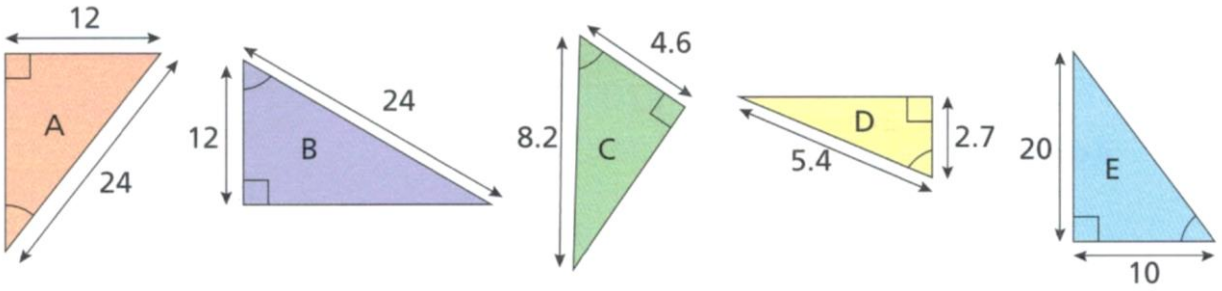
$x = 6$ cm

c



$x = 8.4$ cm

5 Use the fact that $\cos 60° = \frac{1}{2}$ to determine in which of these triangles the angle is 60°.



Triangle B and D.

6 A right-angled triangle has cosine ratio of $\frac{1}{2}$. Every side of the triangle is made 3 times longer. What is the new cosine ratio? Explain your answer using a diagram if necessary.

It is still $\frac{1}{2}$, as the triangles are similar, the ratios are equal.

7 A right-angled triangle has an interior angle of $45°$. Is it always isosceles? Explain by using the tangent ratio.

$\tan 45° = \frac{O}{A} = 1$

Since $\frac{O}{A} = 1$, $O = A$

Two sides of a triangle are equal, so it is isosceles.

8 Explain why it is impossible to have a right-angled triangle with these properties.

a $\tan\theta = -1$

$\frac{O}{A} = -1$, this implies either O or A is negative, but a length can't be negative.

b $\sin\theta = 1$

$\frac{O}{H} = 1$, so $O = H$, impossible the hypotenuse is always the longest side.

c $\cos\theta = 0$

$\frac{A}{H} = 0$, so $A = 0$, but length can't be zero.

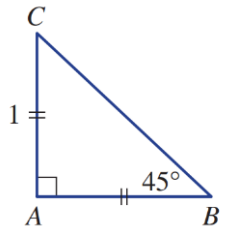
d $\sin\theta > 1$ or $\cos\theta > 1$

$\frac{O}{H} > 1$, $O > H$ or $\frac{A}{H} > 1, A > H$

Impossible as hypotenuse is always the longest side.

9 For the triangle shown,

a use Pythagoras' theorem to find the exact length BC.



$\sqrt{2}$

b use your result from part **a** to write down the exact values of:

a $\sin 45°$

$\frac{1}{\sqrt{2}}$

b $\cos 45°$

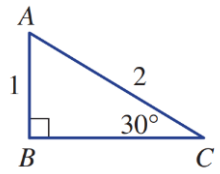
$\frac{1}{\sqrt{2}}$

c $\tan 45°$

1

10 For the triangle shown,

a use Pythagoras' theorem to find the exact length BC.



$\sqrt{3}$

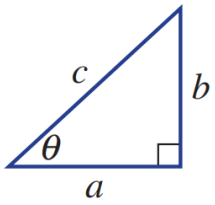
b use your result from part **a** to write down the exact values of:

a $\sin 30°$	**b** $\cos 30°$	**c** $\tan 30°$
$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}}$
d $\sin 60°$	**e** $\cos 60°$	**f** $\tan 60°$
$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$

11 Evaluate $\frac{\sin 20°}{\cos 20°}$. How does this compare to $\tan 20°$?

They are equal.

For this triangle with side lengths a, b and c , find an expression for:



a $\sin\theta$	**b** $\cos\theta$	**c** $\tan\theta$	**d** $\frac{\sin\theta}{\cos\theta}$
$\frac{b}{c}$	$\frac{a}{c}$	$\frac{b}{a}$	$\frac{\left(\frac{b}{c}\right)}{\left(\frac{a}{c}\right)}$ $= \frac{b}{c} \div \frac{a}{c}$ $= \frac{b}{c} \times \frac{c}{a}$ $= \frac{b}{a}$

Evaluating a Trigonometric Ratio

ⓘ Trigonometric Ratios are Stored in your Calculator

- Trigonometric ratios for a given angle are constant in all right-angled triangles.
- Calculators store these ratios.
- Knowing an angle lets you find the ratio between two sides.

🔍 Calculate the value of a trigonometric ratio

$\cos 84°$ cos(84)

≈ 0.1045

$\tan 15°$ tan(15)

≈ 0.2679

What does $\cos 84°$ mean?

Answer using these words: *right-angled triangle, adjacent, hypotenuse, ratio*.

$\cos 84°$ *refers to the ..... of ..... divided by ..... in a .....-angled ..... with an angle of .....*

Evaluate the following to 4 decimal places.

a	$\sin 12°$	**b**	$\cos 12°$	**c**	$\tan 42°$
	0.2079		0.9781		0.9004
d	$7\cos 29°$	**e**	$3\tan 56°$	**f**	$14\cos 12°$
	6.1223		4.4477		13.6941
g	$12\sin 19.3°$	**h**	$\frac{\tan 56°}{3}$	**i**	$\frac{4\cos 16°}{5}$
	3.9662		0.4942		0.7690

1 Use a calculator to evaluate the following correct to 4 decimal places.

a	$\sin 20° =$	**b**	$\cos 37° =$	**c**	$\tan 64° =$
	0.3420		0.7986		2.0503
d	$\sin 47° =$	**e**	$\cos 84° =$	**f**	$\tan 14° =$
	0.7314		0.1045		0.2493
g	$\cos 13° =$	**h**	$\tan 47° =$	**i**	$\sin 36° =$
	0.9744		1.0724		0.5878
j	$12 \sin 13° =$	**k**	$6 \tan 82.3° =$	**l**	$9 \sin 31.3° =$
	2.6994		44.3770		4.6757
m	$13. \sin 31° =$	**n**	$150 \cos 18.5° =$	**o**	$0.5 \tan 51.63° =$
	6.6955		142.2485		0.6315
p	$\frac{9}{\sin 57°} =$	**q**	$\frac{28}{\cos 74°} =$	**r**	$\frac{53.2}{\tan 42°} =$
	10.7313		101.5827		59.0846
s	$\frac{4.2}{\tan 31°} =$	**t**	$\frac{0.64}{\sin 18.2°} =$	**u**	$\frac{291}{\cos 3.45°} =$
	6.9900		2.0491		291.5283

Finding the Value of an Angle

ⓘ Finding the Value of an Angle

- Given an angle, you can use trigonometry to find the ratio of two sides of a triangle.
- The reverse is also true: given a trigonometric ratio, you can find the angle.
- To do so, use the **inverse** trigonometric function:

Inverse Sine	Inverse Cosine	Inverse Tangent
$\sin\theta = \frac{O}{H}$ $\theta = \sin^{-1}\left(\frac{O}{H}\right)$	$\cos\theta = \frac{A}{H}$ $\theta = \cos^{-1}\left(\frac{A}{H}\right)$	$\tan\theta = \frac{O}{A}$ $\theta = \tan^{-1}\left(\frac{O}{A}\right)$
$\sin^{-1}(0.4369)$ is read "sine inverse 0.4369" and means "the angle with a sine ratio of 0.4369"		

🔍 Find the value of an angle

$\sin\theta = 0.6314$

$\theta = \sin^{-1}(0.6314)$ SHIFT sin

$\approx 39°$

$\cos\theta = \frac{8}{10}$

$\theta = \cos^{-1}\left(\frac{8}{10}\right)$ SHIFT COS

$\approx 37°$

What does $\sin^{-1}(0.6314)$ mean?

Answer using these words: *right-angled triangle, opposite, hypotenuse, ratio, angle, inverse*

$\sin^{-1}(0.6314)$*, which is read as "sine ..... 0.6314", is used to find the ..... in a .....-..... ..... where the ..... of ..... to ..... is 0.6314.*

How do you know when to apply the inverse trigonometric ratio?

We apply the inverse trigonometric ratio when trying to find an .....

How do you input $\sin^{-1}(0.6314)$ in a calculator?

First press the ..... button, then press the ..... button.

Calculate the value of θ in these equations. Round your answer to 4 decimal places.

a $\sin\theta = 0.4371$

$\theta = \sin^{-1}(\quad)$

$\approx$

25.9190°

b $\cos\theta = 0.1445$

$\theta = \cos^{-1}(\quad)$

$\approx$

81.6917°

c $\tan\theta = \frac{5}{4}$

$\theta =$

$\approx$

51.3402°

d $\cos\theta = \frac{7}{12}$

54.3147°

e $\tan\theta = 1$

45°

f $\sin\theta = \frac{1}{2}$

30°

1 Use a calculator to solve for θ. Answer to the nearest whole.

a $\sin\theta = 0.5$

$\theta = \sin^{-1}(0.5)$

$\theta =$ .....

30°

b $\cos\theta = 0.5$

$\theta = \cos^{-1}(\quad)$

$\theta =$ .....

60°

c $\tan\theta = 1$

$\theta =$ .....

$\theta =$ .....

45°

d $\cos\theta = 0.8660$

30°

e $\sin\theta = 0.7071$

45°

f $\tan\theta = 0.5774$

30°

g $\sin\theta = 1$

90°

h $\tan\theta = 1.192$

50°

i $\cos\theta = 0$

90°

2 Use a calculator to solve for θ. Answer to 4 decimal places.

a $\sin\theta = \frac{4}{7}$

$\theta = \sin^{-1}\left(\frac{4}{7}\right)$

$\theta =$ .....

34.8499°

b $\cos\theta = \frac{1}{4}$

75.5225°

c $\tan\theta = \frac{3}{5}$

30.9638°

d $\cos\theta = \frac{4}{5}$

36.8699°

e $\sin\theta = \frac{1}{3}$

19.4712°°

f $\tan\theta = \frac{8}{5}$

57.9946

g $\tan\theta = \frac{7}{3}$

66.8014°

h $\cos\theta = \frac{9}{13}$

46.1869°

i $\sin\theta = \frac{15}{42}$

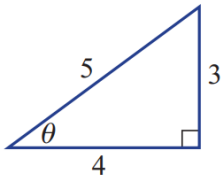
20.9248°

Finding Unknown Sides Part 1

Review

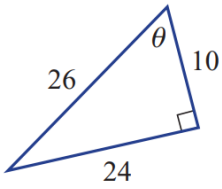
1 Write the three trigonometric ratios, $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$ $\tan\theta = \frac{O}{A}$

a



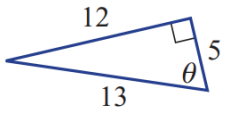
$\sin\theta = \frac{\square}{\square}$ $\cos\theta = \frac{\square}{\square}$ $\tan\theta = \frac{\square}{\square}$

b



$\sin\theta = \frac{\square}{\square}$ $\cos\theta = \frac{\square}{\square}$ $\tan\theta = \frac{\square}{\square}$

c



$\sin\theta = \frac{\square}{\square}$ $\cos\theta = \frac{\square}{\square}$ $\tan\theta = \frac{\square}{\square}$

2 Solve these equations.

a $\frac{x}{6} = 7$

b $\frac{x}{4} = 5$

c $\frac{x}{7} = 24$

3 Match the expressions to their value.

$10 \sin 27°$	62.7
$\frac{58.5}{\tan 43°}$	4.6
$\frac{14.13}{\cos 77°}$	353.6
$11.4 \tan 22°$	4.5
$58 \cos 39°$	45.1
$\frac{155}{\sin 26°}$	62.6
$8.8 \tan 82°$	62.8

Trigonometric Equations

- A **trigonometric ratio** is a number.
- Solve trigonometric equations by treating ratios as numbers.

$$\sin 30° = \frac{x}{7} \qquad \text{is the same as} \qquad 0.5 = \frac{x}{7}$$

Example Solve trigonometric equations: unknown in numerator

$$\cos 60° = \frac{x}{5}$$
$$\times 5 \quad \times 5$$
$$5\cos 60° = x$$
$$2.5 = x$$

Guided Practice

Solve these equations. Round your answer to 2 decimal places.

a

$$\cos 24° = \frac{x}{10}$$

$x = 9.14$

b

$$\sin 29° = \frac{x}{7}$$

$x = 3.39$

c

$$\tan 38° = \frac{x}{13}$$

$x = 10.16$

d

$$\sin 23° = \frac{x}{0.7}$$

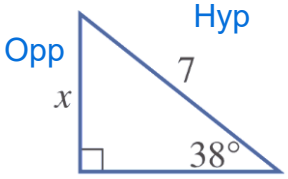
$x = 0.27$

Solving for an Unknown Side

1. Label the side you want to find and given side of the right-angled triangle.
2. Determine the trigonometric ratio that relates these sides.
3. Write the trigonometric equation.
4. Substitute known values into the equation.
5. Solve the equation.

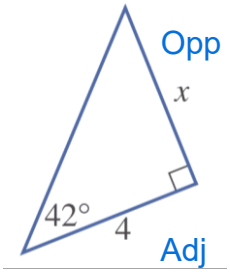
Example Calculate an unknown side using trigonometry; unknown in numerator

Calculate the value of x in these right-angled triangles.



1. Label
2. sin
3. $\sin\theta = \frac{O}{H}$
4. $\sin 38° = \frac{x}{7}$
5. $7\sin 38° = x$

$x \approx 4.31$



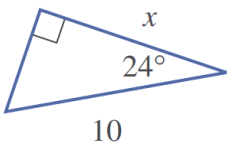
1. Label
2. tan
3. $\tan\theta = \frac{O}{A}$
4. $\tan 42° = \frac{x}{4}$
5. $4\tan 42° = x$

$x \approx 3.60$

Guided Practice

Calculate the value of x in these right-angled triangles. Round your answer to 2 d.p.

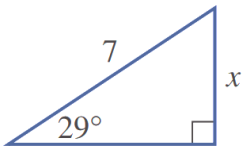
a



1. Label
2.
3.
4.
5.

$x = 9.14$

b



1. Label
2.
3.
4.
5.

$x = 3.39$

1 Solve these equations to find the value of x correct to 2 decimal places.

a $\sin 50° = \frac{x}{4}$

$4 \times \sin 50° = x$

$\ldots\ldots\ldots\ldots = x$

3.06

b $\tan 81° = \frac{x}{3}$

$\ldots \times \tan 81° = x$

$\ldots\ldots\ldots\ldots = x$

18.94

c $\cos 33 = \frac{x}{6}$

$\ldots \times \ldots\ldots = \ldots\ldots$

$\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots$

5.03

d $\cos 75° = \frac{x}{3.5}$

0.91

e $\sin 24° = \frac{x}{4.2}$

1.71

f $\tan 42 = \frac{x}{10}$

9.00

g $\frac{x}{7.1} = \tan 18.4°$

2.36

h $\frac{x}{5.3} = \sin 64.7°$

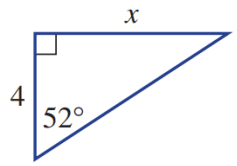
4.79

i $\frac{x}{12.6} = \cos 52.9°$

7.60

2 Determine the two sides given (O, A or H), and hence which trigonometric ratio (sin, cos, or tan) should be used to solve for x.

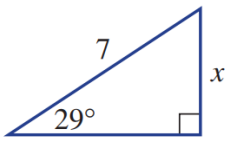
a



○ sin ○ cos ○ tan

tan

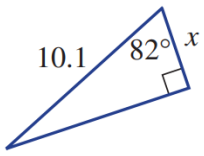
b



○ sin ○ cos ○ tan

sin

c

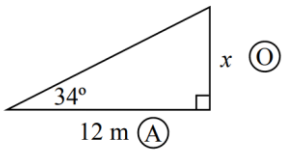


○ sin ○ cos ○ tan

cos

3 Find the side length by setting out your working as shown. Round your answer to 1 d.p.

a



$\tan 34 = \frac{\square}{\square}$ O / A

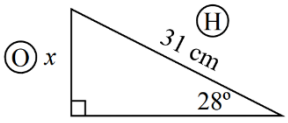
$\times 12 \quad \times 12$

$x = \square \tan \square$

$x = \square$ m 1 d.p.

8.1 m

b



$\sin 28 = \frac{\square}{\square}$ O / H

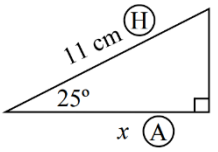
$\times 31 \quad \times 31$

$x = \square \sin \square$

$x = \square$ cm 1 d.p.

4.6 m

c



$\cos 25 = \frac{\square}{\square}$ A / H

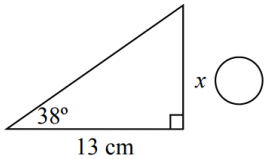
$\times 11 \quad \times 11$

$x = \square \cos \square$

$x = \square$ cm 1 d.p.

10.0 m

d



$\tan \square = \frac{\square}{\square}$ O / A

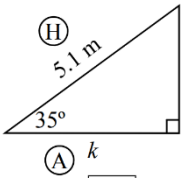
$\times 13 \quad \times 13$

$x = \square \tan \square$

$x = \square$ cm 1 d.p.

10.2 cm

e



$\square = \frac{\square}{\square}$ A / H

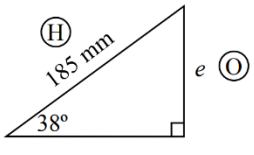
$\times 5.1 \quad \times 5.1$

$k = \square$

$k = \square$ m 1 d.p.

4.4 m

f



$\square = \frac{\square}{\square}$ O / H

$\times 185 \quad \times 185$

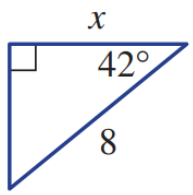
$e = \square$

$e = \square$ mm 1 d.p.

113.9 mm

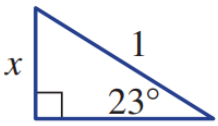
4 Calculate the value of x correct to 2 decimal places using the sine, cosine, or tangent ratios.

a



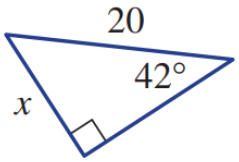
5.95

b



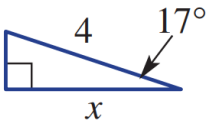
0.39

c



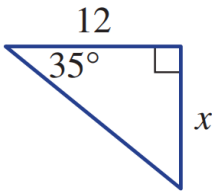
13.38

d



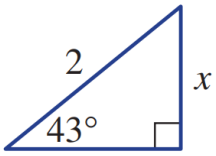
3.83

e



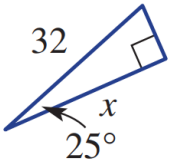
8.40

f



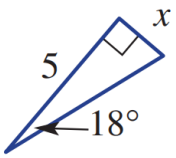
1.36

g



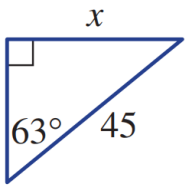
29.00

h



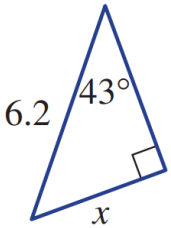
1.62

i



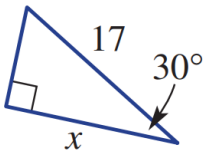
40.10

j



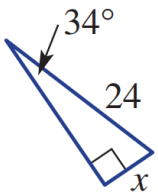
4.23

k



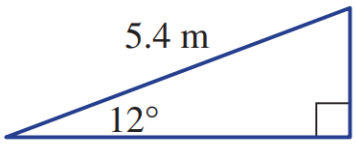
14.72

l



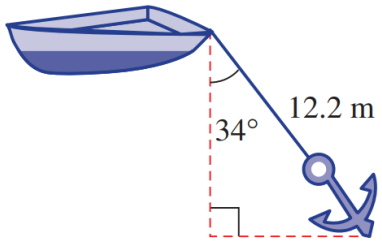
13.42

5 Amy walks 5.4 m up a ramp that is inclined at 12° to the horizontal.
How high (correct to 2 decimal places) is she above her starting point?



1.12 m

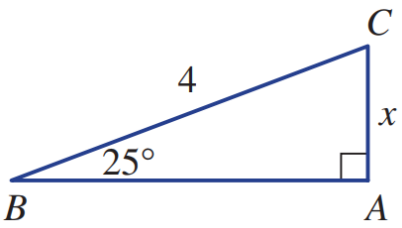
6 One end of a 12.2 m rope is tied to a boat. The other end is tied to an anchor, which is holding the boat steady in the water. If the anchor is making an angle of 34° with the vertical, how deep is the water? Give your answer correct to 2 decimal places.



10.11 m

7 Consider the right-angled triangle on the right.

a Find angle $\angle C$



65°

b Calculate x using the sine ratio.

1.69

c Calculate x using the cosine ratio.

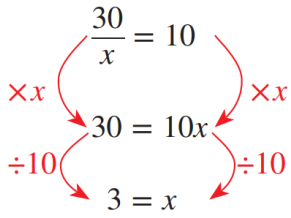
1.69

Finding Unknown Sides Part 2

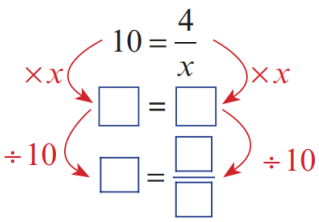
Review

1 To solve an equation with an unknown as the denominator we can first multiply both sides by that unknown. Solve these equations.

a



b



c $\frac{4}{x} = 2$

d $\frac{20}{x} = 4$

e $\frac{15}{x} = 5$

f $2 = \frac{12}{x}$

Example Solve trigonometric equations: unknown in denominator

$$\cos 60° = \frac{5}{x}$$
$$\times x \quad \times x$$
$$x \cos 60° = 5$$
$$\div \cos 60° \quad \div \cos 60°$$
$$x = \frac{5}{\cos 60°}$$
$$= 10$$

Guided Practice

Solve these equations. Round your answer to 2 decimal places.

a $\cos 47° = \frac{26}{x}$

$x = 38.12$

b $\sin 53° = \frac{26}{x}$

$x = 32.56$

c $\tan 44° = \frac{15}{x}$

$x = 15.53$

d $\cos 49° = \frac{6}{x}$

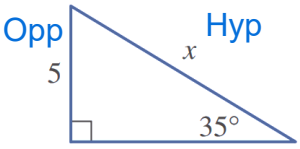
$x = 9.15$

Solving for an Unknown Side

1. Label sides.
2. Determine trig ratio.
3. Write trig equation.
4. Substitute known values.
5. Solve equation.

Example Calculate an unknown side using trigonometry; unknown in numerator

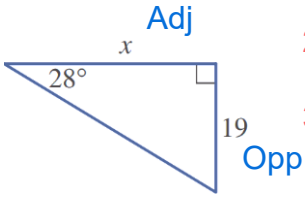
Calculate the value of x in these right-angled triangles.



1. label
2. sin
3. $\sin\theta = \frac{O}{H}$
4. $\sin 35° = \frac{5}{x}$
5. $x \sin 35° = 5$

$$x = \frac{5}{\sin 35°}$$

$$\approx 8.72$$



1. label
2. tan
3. $\tan\theta = \frac{O}{A}$
4. $\tan 28° = \frac{19}{x}$
5. $x \tan 28° = 19$

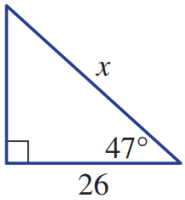
$$x = \frac{19}{\tan 28°}$$

$$\approx 35.73$$

Guided Practice

Calculate the value of x in these right-angled triangles. Round your answer to 2 d.p.

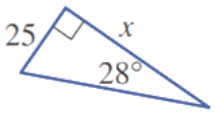
a



1. Label
2.
3.
4.
5.

$x = 38.12$

b



$x = 47.02$

1 Solve the equation to find the value of x correct to 2 decimal places.

a $\cos 43° = \frac{3}{x}$

$x \cos 43° = 3$

$x = \frac{3}{\cos 43°}$

$= \ldots\ldots\ldots$

4.10

b $\sin 36° = \frac{4}{x}$

$x \sin 36° = 4$

$x = ____$

$= \ldots\ldots\ldots$

6.81

c $\cos 33 = \frac{6}{x}$

7.15

2 Emma noticed a shortcut when solving these equations:



Instead of writing:

$2 = \frac{3}{x}$

$2x = 3$

$x = \frac{3}{2}$

We can skip a step:

$2 = \frac{3}{x}$

swap

$x = \frac{3}{2}$

Solve these equations using the shortcut.

a $\tan 64° = \frac{2}{x}$

$x = \frac{2}{\tan 64°}$

$x =$

0.98

b $\cos 67° = \frac{5}{x}$

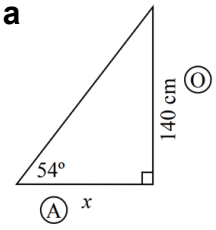
12.80

c $\sin 12° = \frac{3}{x}$

14.43

3 Find the side length by setting out your working as shown. Round your answer to 1 d.p.

a



$\tan 54 = \frac{\square}{\square}$ (O / A)

swap swap

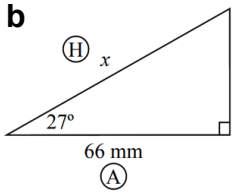
$x = \frac{\square}{\tan \square}$

$x = \square$ cm

1 d.p.

101.7 cm

b



$\cos 27 = \frac{\square}{\square}$ (H / A)

swap swap

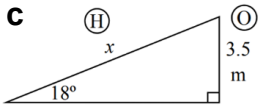
$x = \frac{\square}{\cos \square}$

$x = \square$ mm

1 d.p.

74.1 mm

c



$\sin 18 = \frac{\square}{\square}$ (O / H)

swap swap

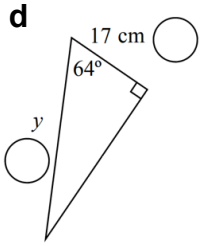
$x = \frac{\square}{\sin \square}$

$x = \square$ m

1 d.p.

11.3 m

d



$\square = \frac{\square}{\square}$

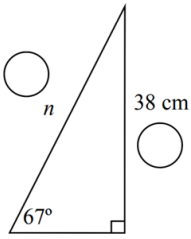
swap swap

$y = \frac{\square}{\square}$

$y = \square$ cm

18.9 cm

e



$\square = \frac{\square}{\square}$

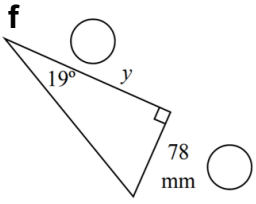
swap swap

$n = \frac{\square}{\square}$

$n = \square$ cm

41.3 cm

f



$\square = \frac{\square}{\square}$

swap swap

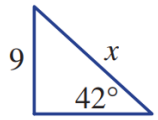
$y = \frac{\square}{\square}$

$y = \square$ mm

226.5 mm

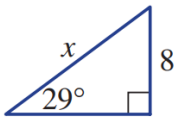
4 Calculate the value of x correct to 2 decimal places.

a



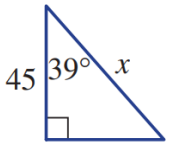
13.45

b



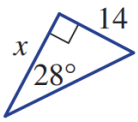
16.50

c



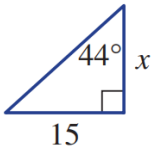
57.90

d



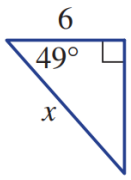
26.33

e



15.53

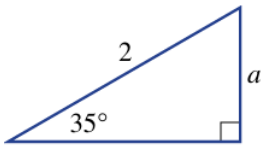
f



9.15

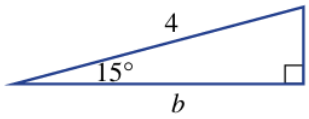
5 Calculate the value of the pronumeral, to 2 decimal places.

a



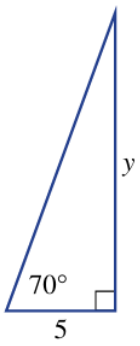
1.15

b



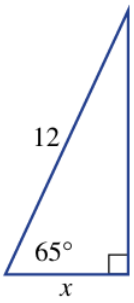
3.86

c



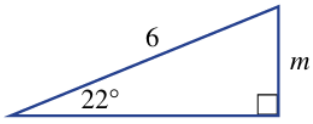
13.74

d



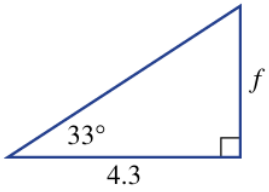
5.07

e



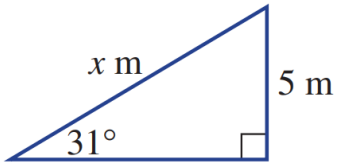
2.25

f



2.79

6 Solve for x.



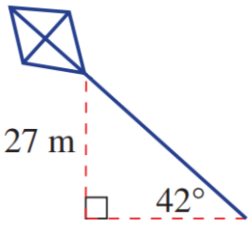
Two students provide different answers in the test. Which student is more accurate? What mistake have they made?

A $x = \dfrac{5}{\sin 31°} = \dfrac{5}{0.52} = 9.62$

B $x = \dfrac{5}{\sin 31°} = 9.71$

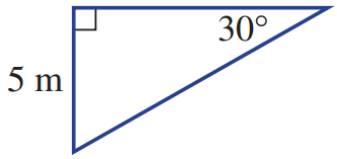
Student B is more accurate. Student A rounded too early.

7 A kite is flying at a height of 27 m above the anchor point. If the string is inclined at 42° to the horizontal, find the length of the string correct to the nearest metre.



40 m

8 Find the perimeter of this triangle, correct to 1 d.p.



23.7 m

9 Find the length of the side labelled x.

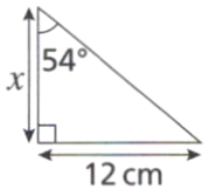
a by using the 54° angle

$\tan 54° = \dfrac{12}{x}$

$x = \dfrac{12}{\tan 54°}$

$= 8.72$ cm

b By using the unmarked angle in the triangle.



Missing angle is $180 - 90 - 54 = 36°$

$\tan 36° = \dfrac{x}{12}$

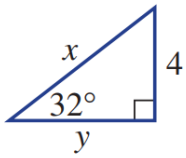
$x = 12 \tan 36°$

$= 8.72$ cm

Which method do you prefer?

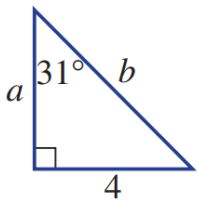
10 Find the value of each pronumeral correct to 1 decimal place.

a



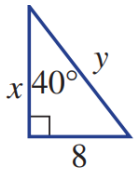
$x = 7.5, y = 6.4$

b



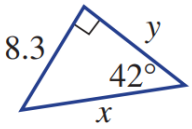
$a = 6.7, b = 7.8$

c



$x = 9.5, y = 12.4$

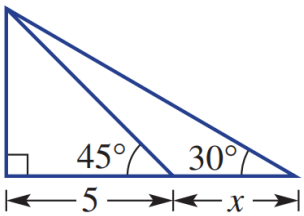
d



$x = 12.4, y = 9.2$

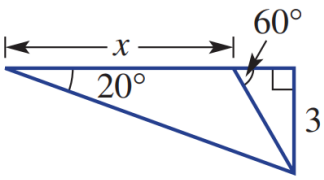
11 Find the value of x, correct to 1 decimal place, in these triangles.

a



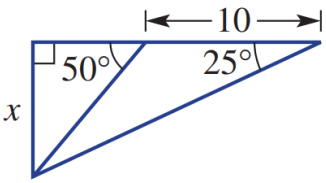
3.7

b



6.5

c



7.7

Finding Unknown Angles

Review

1 Solve these equations.

a $\sin\theta = 0.4$

$\theta = \sin^{-1}(\quad)$

$=$

b $\cos\theta = \frac{4}{5}$

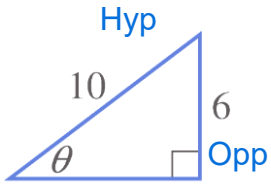
c $\tan\theta = 32.4$

Solving for an Unknown Side

1. Label sides.
2. Determine trig ratio.
3. Write trig equation.
4. Substitute known values.
5. Solve equation.

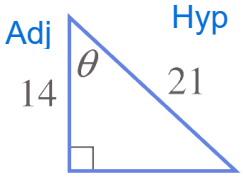
Example Calculate an unknown side using trigonometry; unknown in numerator

Calculate the unknown angle in these right-angled triangles.



1. Label
2. sin
3. $\sin\theta = \frac{O}{H}$
4. $\sin\theta = \frac{6}{10}$
5. $\theta = \sin^{-1}\left(\frac{6}{10}\right)$

   $\theta \approx 36.87°$



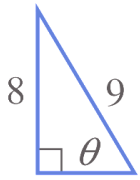
1. Label
2. cos
3. $\cos\theta = \frac{A}{H}$
4. $\cos\theta = \frac{14}{21}$
5. $\theta = \cos^{-1}\left(\frac{14}{21}\right)$

   $\theta \approx 48.19°$

Guided Practice

Calculate the value of θ in these right-angled triangles. Round your answer to 2 d.p.

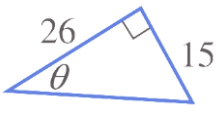
a



1. Label
2.
3.
4.
5.

$\theta = 62.73°$

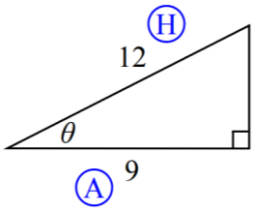
b



$\theta = 29.98°$

1 Calculate the value of θ to the nearest degree.

a



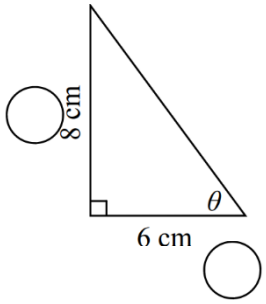
$\cos\theta = \frac{9}{12}$

$\theta = \cos^{-1}\left(\frac{9}{12}\right)$

$\theta =$

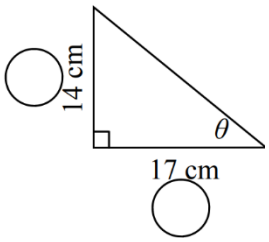
41°

b



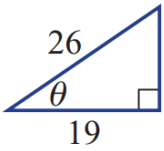
53°

c



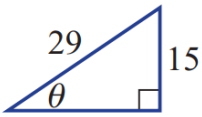
39°

d



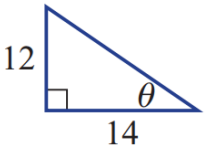
43°

e



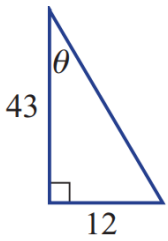
31°

f



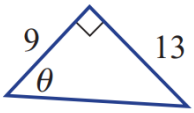
41°

g



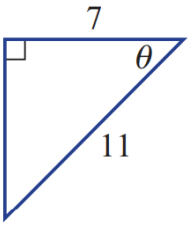
16°

h



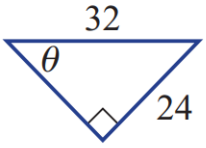
55°

i



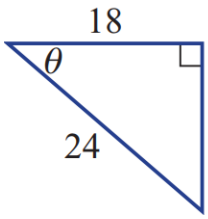
50°

j



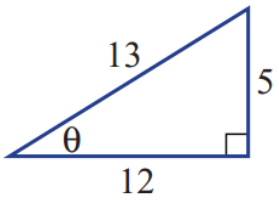
49°

k



41°

2 Refer to the diagram to complete the following, answer to 2 decimal places.



a $\sin\theta = \frac{\square}{\square}$

$\theta =$

22.62°

b $\cos\theta = \frac{\square}{\square}$

$\theta =$

22.62°

c $\tan\theta = \frac{\square}{\square}$

$\theta =$

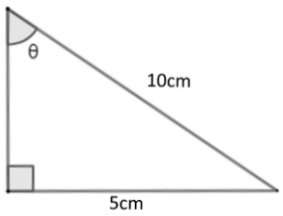
22.62°

What do you notice? Does it make sense?

The answer for all three is 22.62°. This makes sense as you are finding the same angle in each case.

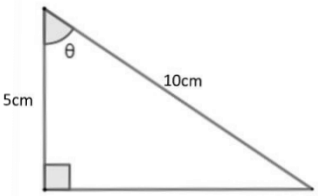
3 Solve for the unknown angle.

a



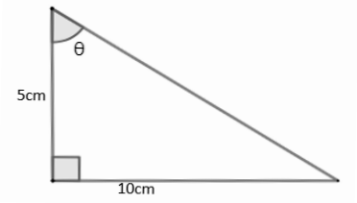
30°

b



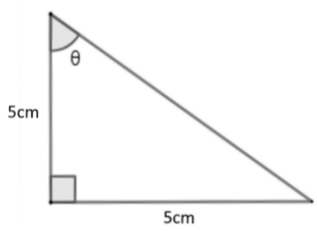
60°

c



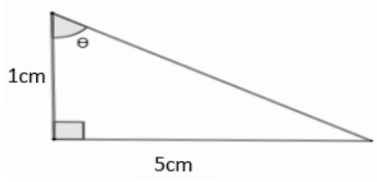
63.4°

d



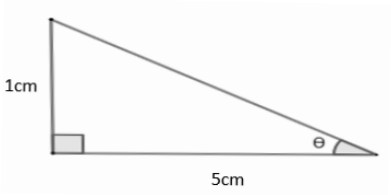
45°

e



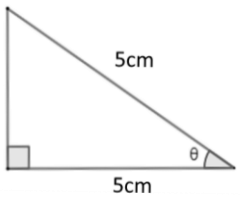
78.7°

f



11.3°

4 Solve for θ



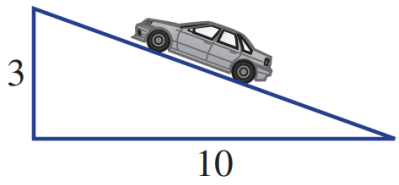
0°

Explain why this triangle is impossible.

It's impossible for a triangle to have an internal angle of zero.

Also, the triangle's hypotenuse and adjacent are equal, which is impossible as the hypotenuse must be the longest side.

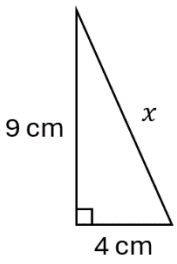
5 A road rises at a grade of 3 in 10.
Find the angle (to the nearest degree) the road makes with the horizontal.



17°

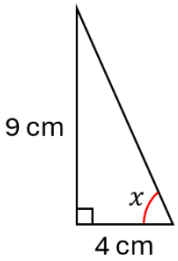
6 Find the value of x. Answer to 2 decimal places.

a



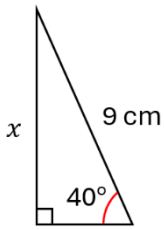
$\sqrt{97} = 9.85$ cm

b



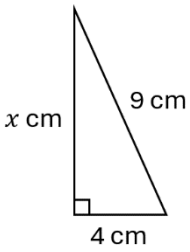
66.04°

c



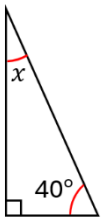
5.79 cm

d



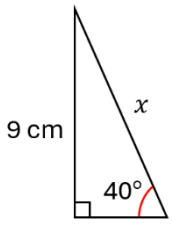
$\sqrt{65} = 8.06$ cm

e



50°

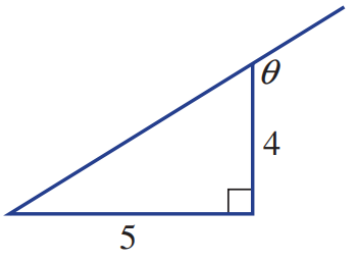
f



14.00 cm

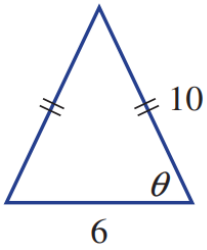
7 Calculate the value of θ in these diagrams. Answer to 1 d.p.

a



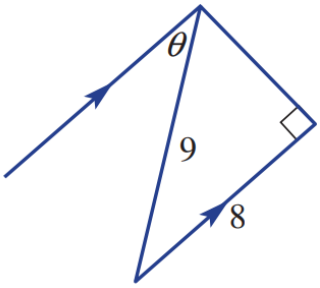
128.7°

b



72.5°

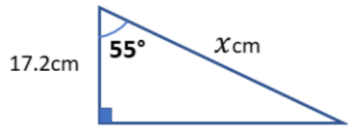
c



27.3°

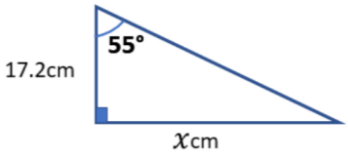
12 Solve for the unknown side. Answer to 1 d.p.

a



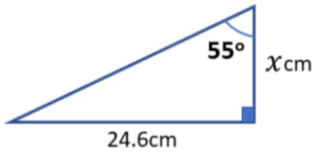
30.0 cm

b



24.6 cm

c



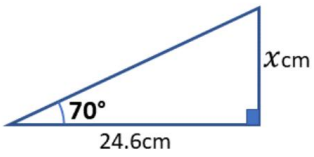
17.2 cm

d



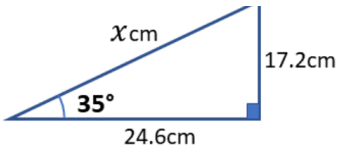
17.2 cm

e



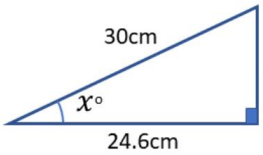
67.6 cm

f



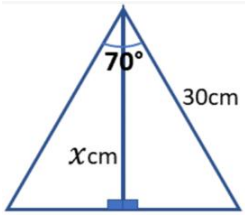
30.0 cm

g



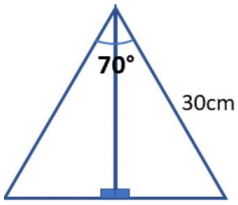
35°

h



24.6 cm

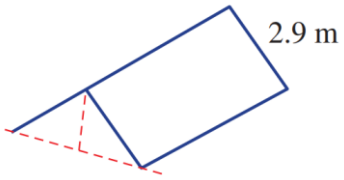
i



Area = ?

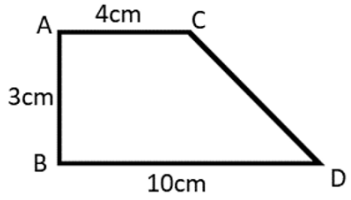
422.9 cm^2

8 A tent with a simple A-frame design is set up as a shelter. The breadth of half of the tent is 2.9 metres, as shown. Find the angle to the ground that the sides of the tent make if the height at the middle of the shelter is 1.5 metres. Round your answer to the nearest 0.1 of a degree.



31.1°

9 Consider the following shape; answer all questions to 1 d.p,



a Calculate the area of ΔBCD

15 cm²

b Calculate the perimeter of the trapezium.

23.7 cm

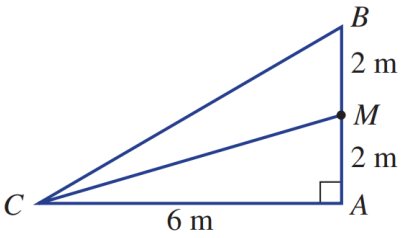
c Find the size of $\angle CDB$ to 1 d.p.

26.6°

d Find the size of $\angle ACD$ to 1 d.p.

153.4°

10 M is the midpoint of AB. Determine whether the angle $\angle ACM$ is exactly half of angle ACB.



∠ACM = 18.4° ∠ACB = 33.7° , no it is not half

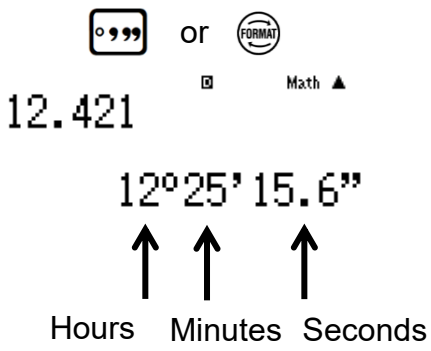
Degrees Minutes Seconds

ⓘ Sexagesimal System

- Our number system is called the **decimal** system based on tens.
  - Ten ones are ten. Ten tens are a hundred. Ten hundreds is a thousand…
- Angles use the **sexagesimal** system, based on sixties.
  - Sixty seconds are a minute. Sixty minutes are a degree.

Degree °	$\frac{1}{360}$ of a turn
Minute ′	$\frac{1}{60}$ of a degree
Second ″	$\frac{1}{60}$ of a minute

Convert between decimal form and sexagesimal form by using the buttons:



🔍 **Convert** between sexagesimal form and decimal form

32.5° [3] [2] [•] [5] [=] [°′″]
= 32° 30′

89.25° [8] [9] [•] [2] [5] [=] [°′″]
= 89° 15′

Convert these angles from decimal form to sexagesimal form

a 44.75° = 44°45′	**b** 36.6° = 36°36′	**c** 0.1° = 0°6′
d 77.2° = 77°12′	**e** 0.5° = 0°30′	**f** 126.1552° = 126° 9′ 18.72″

Rounding in Sexagesimal Form

1. Draw a “cut” after the required unit.
2. Circle the **critical unit** right after the cut.
3. Decide to round up to down.
   i. If critical unit is less than 30, round down.
   ii. If critical unit is 30 or more, round up.
4. Round.
   i. rounding down: Remove digits after the cut.
   ii. rounding up: Add 1 to digit before the cut, then remove digits after the cut.

Example Round to the nearest degree and minute

Round $43° 44′ 8.4″$ to the nearest degree.

1. Draw cut	$43\ °\ \vert\ 44\ ′\ 8.4\ ″$
2. Critical digit	$43\ °\ \vert\ 44\ ′\ 8.4\ ″$
3. Decide	*Round up*
4. Round	$44°$

$43° 44′ 8.4″ \approx 44°$

Round $88° 30′ 22″$ to the nearest minute.

1. Draw cut	$88\ °\ 30\ ′\ \vert\ 22\ ″$
2. Critical digit	$88\ °\ 30\ ′\ \vert\ 22\ ″$
3. Decide	*Round down*
4. Round	$88°30′$

$88° 30′ 22″ \approx 88°30′$

Guided Practice

Round these angles to the nearest minute.

a	$51° 59′ 8″$	**b**	$33° 22′ 47″$	**c**	$41° 24′ 47″$
	$51°59′$		$33°23′$		$41°25′$
d	$42.5825°$	**e**	$74.2842°$	**f**	$14.5°$
	$42°35′$		$74°17′$		$14°30′$

Key Ideas

1. There are 60 ..... in 1 degree.
2. There are 60 ..... in 1 minute.
3. When rounding to the nearest degree or minute, we use ..... as the halfway mark.

1 Complete these sentences:

a …… degrees make a full turn, or revolution.

b 60 …… make up one degree.

c 60 …… make up one minute.

2 Convert these angles in decimal form to degrees, minutes, and seconds.

a	25.1° = 25°……′ 25° 6' 0"	**b**	25.2° = 25°……′ 25° 12' 0"	**c**	25.25° = 25°……′ 25° 15' 0"
d	25.5° = 25°……′ 25° 30' 0"	**e**	25.625° = 25°……′……″ 25° 37' 30"	**f**	25.75° = …… 25° 45' 0"

3 Round these angles to the nearest degree.

a	11° 38’ 12°	**b**	55°1 6’ 55°	**c**	49° 46’ 50°
d	70° 28’ 70°	**e**	79° 30’ 80°	**f**	18° 27’ 18°
g	42° 21’ 44” 42°	**h**	44° 35’ 21’ 45°	**i**	29° 21’ 45” 29°

4 Round these angles to the nearest minute.

a	43° 52’ 11.71” 43° 52'	**b**	35° 13’ 24.05” 35° 13'	**c**	42° 59’ 28.76” 42° 59'
d	19° 27’ 11.66” 19° 27'	**e**	76° 49’ 23.05’ 76° 49'	**f**	14° 30’ 55.17” 14° 31'

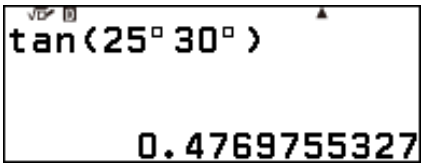
5 Convert these angles in decimal form to sexagesimal form and round to the nearest minute.

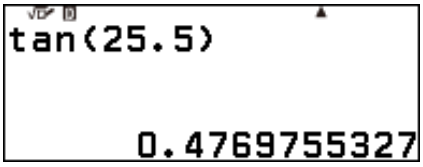
a	25.2875° 25° 17'	**b**	45.16666° 45° 10'	**c**	67.4916° 67° 29'
d	72.4215° 72° 25'	**e**	22.3° 22° 18'	**f**	87.252° 87° 15'

Trigonometry with Sexagesimals

Trigonometry with Sexagesimals

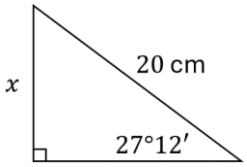
The °′″ button is used to insert angles in terms of degrees and minutes into the calculator



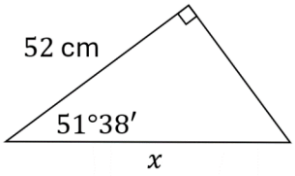


Example Calculate unknown sides when angle is in sexagesimal form

Calculate the value of x. Round your answer to 2 decimal places



$$\sin(27°12') = \frac{x}{20}$$
$$20\sin(27°12') = x$$
$$9.14 \text{ cm} = x$$

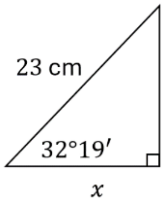


$$\cos(51°38') = \frac{52}{x}$$
$$x\cos(51°38') = 52$$
$$x = \frac{52}{\cos(51°38')}$$
$$= 84.78 \text{ cm}$$

Guided Practice

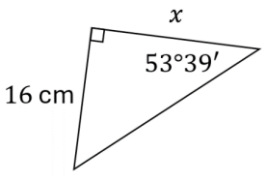
Calculate the value of x. Round to 2 decimal places.

a



19.44 cm

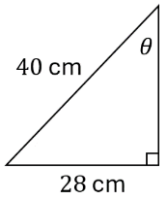
b



$x = 11.77$ cm

Example Calculate an unknown angle and round the result

Calculate the value of θ and round to the nearest minute.



$$\sin\theta = \frac{28}{40}$$

$$\theta = \sin^{-1}\left(\frac{28}{40}\right)$$

$$\theta \approx 44.4270$$

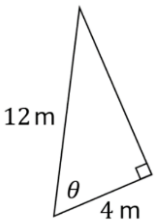
$$\approx 44° 25' 37.21''$$

$$\approx 44° 26'$$

Guided Practice

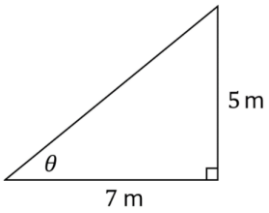
Calculate the value of θ and round to the nearest minute.

a



$\theta = 70°32'$

b

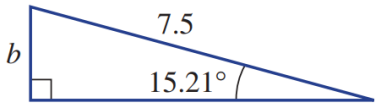
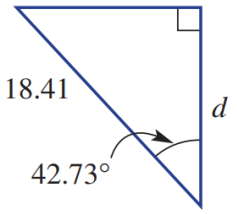
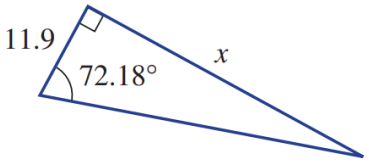
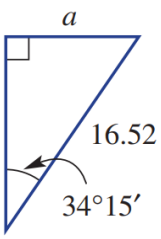
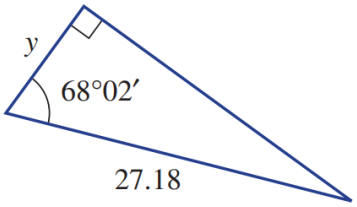
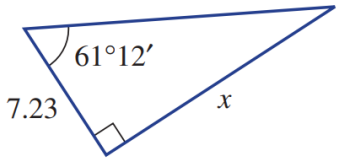


$\theta = 35°32'$

1 Evaluate these trigonometric ratios using the DMS button on the calculator. Answer to 4 d.p.

a cos 89°21′ 0.0113	**b** tan 18°23′ 0.3323	**c** tan 68°23′ 2.5236
d 8.3 tan 58°51′ 13.7320	**e** 15 sin 48.18° 11.1786	**f** 12.3 cos 27°48′ 10.8803

2 Calculate the value of the unknown side in each of the following. Answer to 2 d.p.

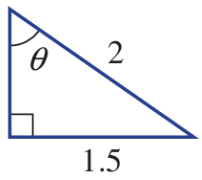
a  1.97	**b**  13.52	**c**  37.02
d  9.30	**e**  10.17	**f**  13.15

3 Calculate θ in each of the following, round to the nearest minute.

a $\sin^{-1}(0.7324)$ 47° 5′	**b** $\cos^{-1}(0.9763)$ 12° 30′	**c** $\tan^{-1}(0.3321)$ 18° 22′
d $\tan^{-1}(1.235)$ 51° 0′	**e** $\sin^{-1}(0.4126)$ 24° 22′	**f** $\cos^{-1}(0.7462)$ 41° 44′
g $\sin\theta = 0.5$ 30° 0′	**h** $\tan\theta = 0.5774$ 30° 0′	**i** $\tan\theta = 1.192$ 50° 0′
j $\sin\theta = \frac{4}{7}$ 34° 51′	**k** $\cos\theta = \frac{4}{5}$ 36° 52′	**l** $\tan\theta = \frac{8}{5}$ 58° 0′

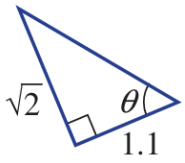
4 Calculate the value of θ in each of the following, round to the nearest minute.

a



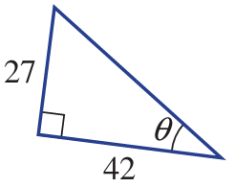
48°35′

b



52°7′

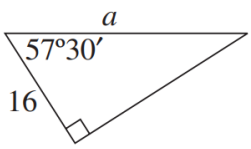
c



32°44′

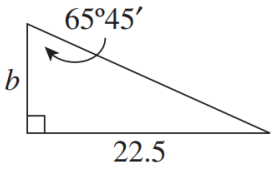
5 Calculate the value of the unknown value, to 2 d.p. for lengths and nearest minute for angles.

a



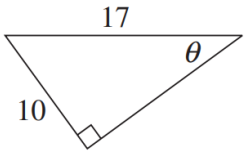
29.78

b



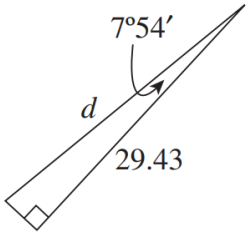
10.14

c



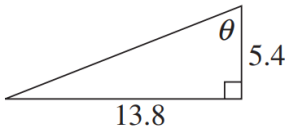
36°2′

d



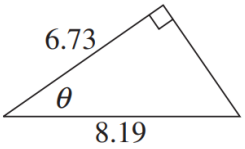
29.71

e



68°38′

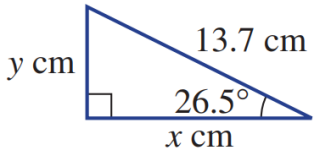
f



34°44′

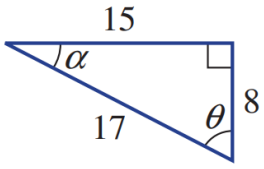
6 Find the values of the pronumerals, to 2 d.p. for lengths and nearest minute for angles.

a



x = 12.26 cm, y = 6.11 cm

b



α = 28°4′ , θ = 61°56′

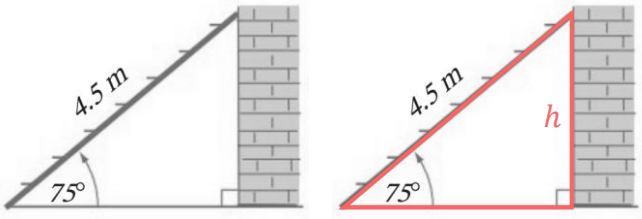
Problem-solving using Trigonometry

Problem-solving using Trigonometry

1. Identify the right-angled triangle.
2. Label the unknown side or angle x or θ, if it isn't already.
3. Solve for the unknown using trigonometry.

Example Solve practical problems involving trigonometry

Find how high the ladder reaches up the wall.



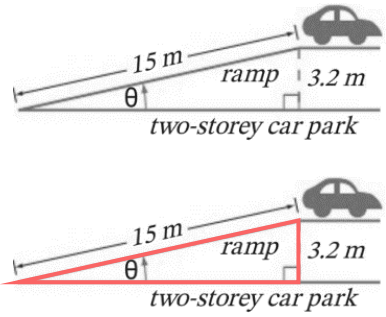
1. Triangle
2. Unknown (h)
3. Solve

$$\sin 75° = \frac{h}{4.5}$$

$$4.5 \sin 75° = h$$

$$x \approx 4.35 \text{ m}$$

Find the angle of the ramp.



1. Triangle
2. Unknown (θ)
3. Solve

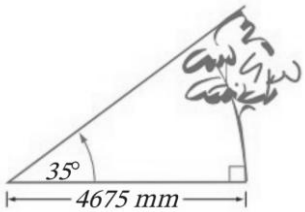
$$\sin \theta = \frac{3.2}{15}$$

$$\theta = \sin^{-1}\left(\frac{3.2}{15}\right)$$

$$\theta \approx 12.32°$$

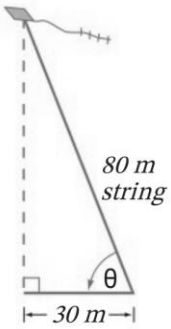
Guided Practice

a Determine the height of this tree.



$x = 3273.47$ mm

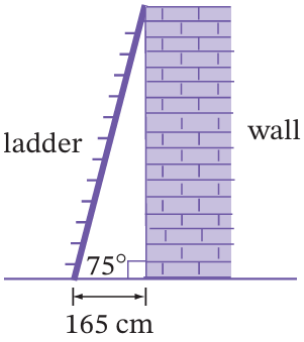
b Find the angle between the kite and the ground.



$\theta = 67.98°$

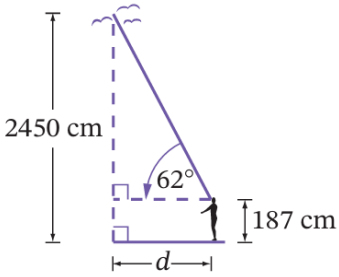
1 For each question, solve for the unknown value.

a Calculate the length of the ladder to the nearest cm.



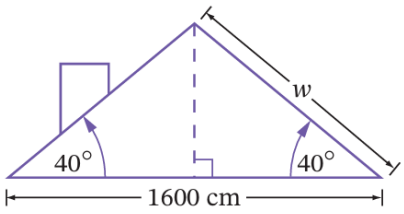
638 cm

b Calculate the horizontal distance of the person from the birds, to 1 d.p.



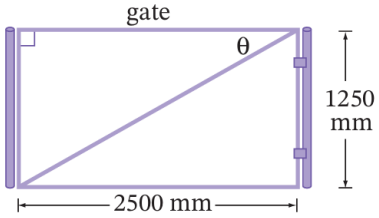
1203.3 cm

c The slant height of the roof, to 1 d.p.



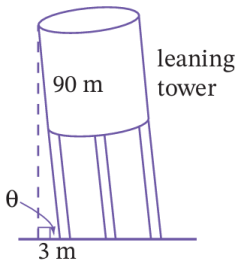
1044.3 cm

d The angle of the gate diagonal, to the nearest minute.



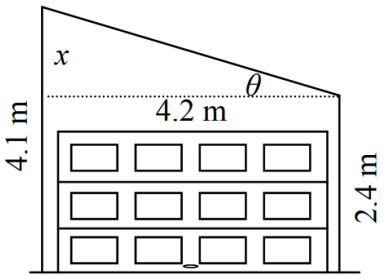
26°34′

e The angle of the leaning water tower, to the nearest minute.



88°5′

2 A garage has a roof shown in the diagram.



a Calculate the height difference between the two walls.

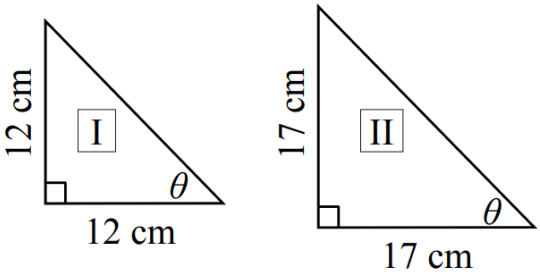
1.7 m

b Find the angle of the roof to the nearest minute.

22°2′

3 Toby constructed two isosceles right triangles.
He says that when the two short sides are the same the angle θ is always $45°$.

a Show he is correct, using trigonometry in the two triangles.



$\tan^1\left(\frac{12}{12}\right) = 45°, \quad \tan^{-1}\left(\frac{17}{17}\right) = 45°$

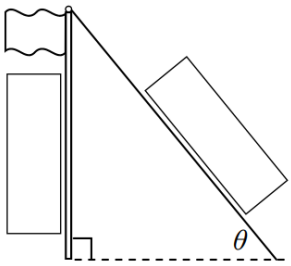
b Show he is correct, using properties of isosceles triangles and angle sum of a triangle.

$\theta + \theta + 90° = 180°$

$2\theta = 90°$

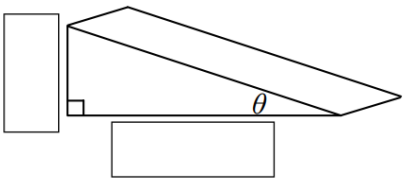
$\theta = 45°$

4 A 4.5 m flagpole has a 5.6 m wire anchored into the ground.
Calculate the angle that the wire makes with the ground to the nearest minute.



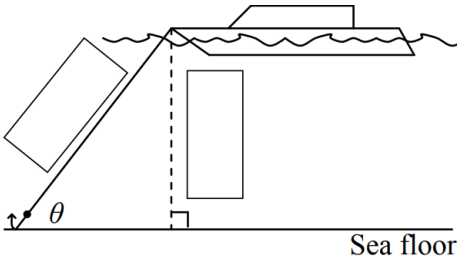
53°28′

5 A triangle skateboard ramp is 3.5 m long and 80 cm high.
Calculate the angle that it makes with the ground to the nearest minute.



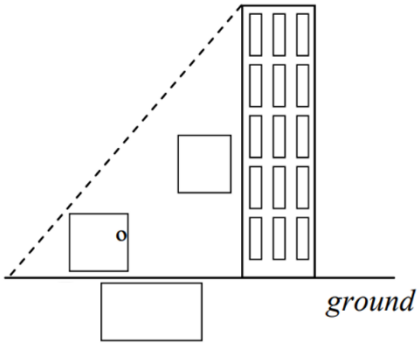
12° 5′

6 A fishing boat is anchored in 65 m deep water. The anchor rope is 90 m long.
Calculate the angle the rope makes with the sea floor, to the nearest minute.



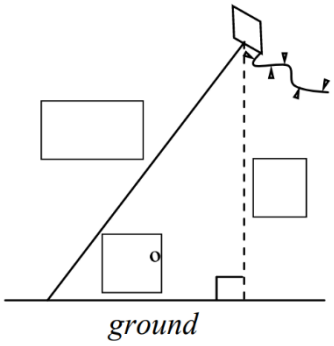
46°14′

7 Jess looks up at a building that is 72 m away. She looks up at an angle of 54°.
Complete the diagram then calculate the height of the building to 1 d.p.



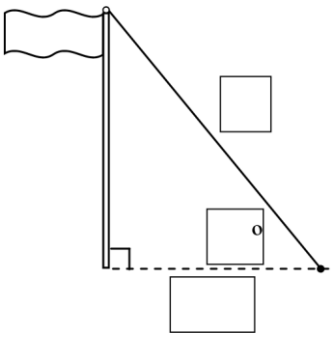
99.1 m

8 A kite string is 55 m long and is fully extended.
The string makes an angle of $64°$ with the ground.
Calculate how high above the ground it is to 1 d.p.



49.4 m

9 A flagpole has a wire that secures it to the ground 6 m from the pole.
The wire makes an angle of $51°$ with the ground.
Calculate the length of the wire.



9.5 m

10 For the following questions, you will need to draw your own diagrams.

a A ladder rests against a wall. The foot of the ladder is 355 cm from the wall and makes an angle of $63°20'$ with the ground.
Calculate the length of the ladder to the nearest cm.

791 cm

b A glider is flying at an altitude (height) of 1.5 km.
To land, it descends at an angle of $18°$ to the ground.
How far must the glider travel before landing? Answer rounded to the nearest 0.1 km.

4.9 km

c The entrance to the school is 60 cm above ground level.
A wheelchair ramp is to be built at an angle of $4°$ to the ground.
How far from the library must the ramp start? Answer to the nearest cm.

858 cm

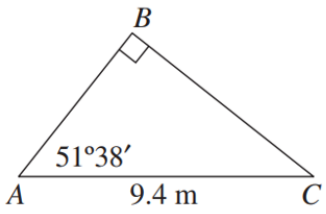
d An aircraft is at an altitude of 770 m. It is 1.5 km horizontally from the airport.
What angle must it descend by to land safely? Answer to the nearest minute.

$27°10'$

e A cannon on level ground shoots at a target, but the cannonball misses by 25 m.
If the target is 1 km away, what angle do they need to adjust the cannon?
Answer to the nearest minute.

$1°26'$

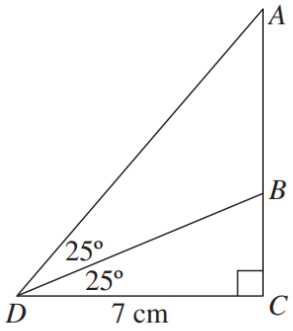
11 A tree snapped into two sections AB and BC in high winds and then fell. The section BA is inclined at $51°38'$ to the horizontal and AC is 9.4 metres long. Find the height of the original tree, in metres correct to one decimal place.



13.2 m

12

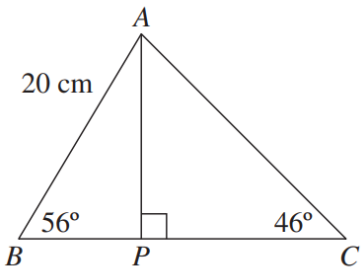
a



Show that $AC = 7\tan 50°$ and $BC = 7\tan 25°$, and hence find the length AB, correct to 1 mm.

51 mm

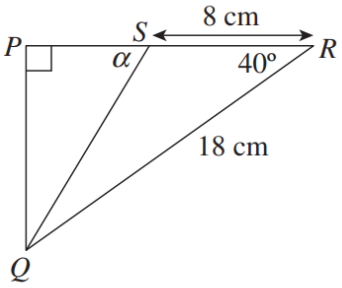
b



Show that $AP = 20\sin 56°$, and hence find the length of PC, giving your answer correct to 1 cm.

16 cm

c

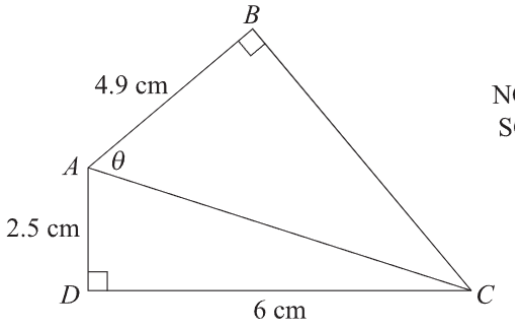


Show that $PR = 18\cos 40°$, find an expression for PQ, and hence find the angle α, correct to the nearest minute.

63°25′

13 **2019 HSC Standard 2 Band 4**

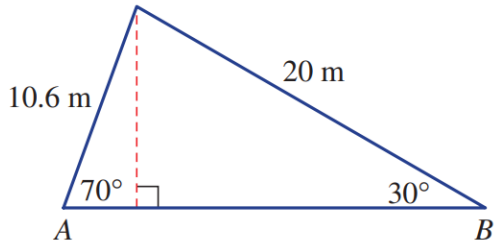
Calculate the size of θ. Answer to the nearest minute.



41°5′

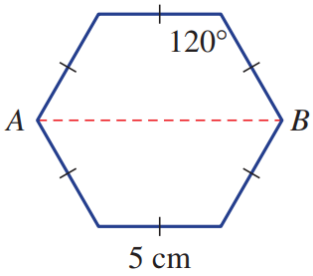
14 Find the length AB in these diagrams. Answer to 2 d.p.

a



20.95 m

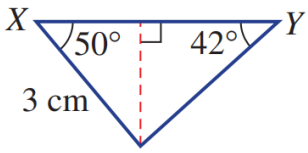
b



10 cm

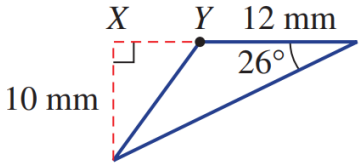
15 Find the length XY in these diagrams. Answer to 1 d.p.

a



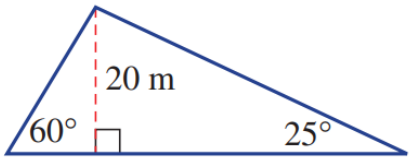
4.5 cm

b



8.5 mm

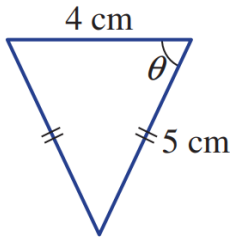
16 Find the perimeter of this triangle.



124.86 m

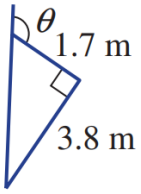
17 Find the angle θ in these diagrams. Answer to the nearest minute.

a



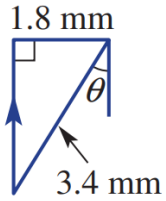
66°25′

b



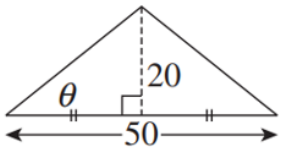
114°6′

c



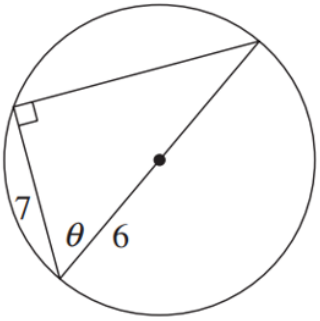
31°58′

d



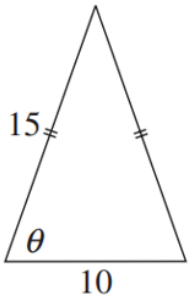
38°40′

e



54°19′

f



70°32′

18 It is given that θ is an acute angle and that $\tan\theta = \frac{\sqrt{5}}{2}$

a Draw a right-angled triangle to show this information.

b Use Pythagoras' theorem to find the length of the unknown side.

$\sqrt{9}$

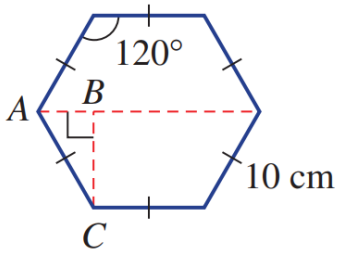
c Hence write the exact values of $\sin\theta$ and $\cos\theta$

$\sin\theta = \frac{\sqrt{5}}{\sqrt{9}}, \cos\theta = \frac{2}{\sqrt{9}}$

d Show that $\sin^2\theta + \cos^2\theta = 1$.

$\left(\frac{\sqrt{5}}{\sqrt{9}}\right)^2 + \left(\frac{2}{\sqrt{9}}\right)^2 = \frac{5}{9} + \frac{4}{9} = 1$

19 Consider a regular hexagon with internal angles of 120° and side lengths of 10 cm.



a Find the length of BC to 2 decimal places.

$\angle BAC = 60°$

$\sin 60° = \frac{BC}{10}$

$BC = 10 \sin 60° = 8.66$ cm

b Find the length of AB.

$\cos 60° = \frac{AB}{10}$

$AB = 10 \cos 60° = 5$ cm

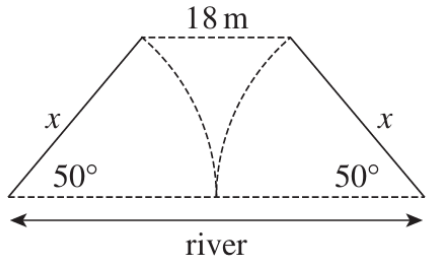
c Find the distance between the parallel sides to 1 decimal place.

$2BC = 20 \sin 60° = 17.3$ cm

d Find the distance between two opposite vertices.

$5 + 10 + 5 = 20$ cm

20 A bridge spans a river, and the two identical sections of the bridge, each of length x metres, can be raised to allow tall boats to pass. When the two sections are fully raised, they are each inclined at 50° to the horizontal, and there is an 18-metre gap between them, as shown in the diagram. Calculate the width of the river in metres, correct to one decimal place.



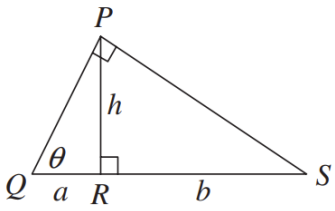
The river is $2x$ m long.

Consider the horizontal projections of the raised bridges: $2x - 2x \cos 50° = 18$

$2x = 50.4$

∴ The river is 50.4 metres long.

21 Consider the diagram.



a Explain why $\angle RPS = \theta$

$\angle QPR = 90° - \theta$, so $\angle RPS = \theta$

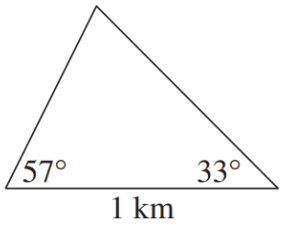
b Find two expressions for $\tan\theta$

$\frac{h}{a}$ and $\frac{b}{h}$

c Hence show $ab = h^2$

$\frac{h}{a} = \frac{b}{h}$, $\therefore h^2 = ab$

22 From the ends of a straight horizontal road 1 km long, a balloon directly above the road is observed to have angles of elevation of 57° and 33° respectively.
Find, correct to the nearest metre, the height of the balloon above the road.
(Hint: create two right-angled triangles by drawing a vertical line)



Draw a vertical line from the apex of the triangle to the base, this represents the height h.

This vertical line splits the base into two sections, which we will label a and b.

$\tan 57° = \frac{h}{a}$, $\tan 33° = \frac{h}{b}$

$a = \frac{h}{\tan 57°}$, $b = \frac{h}{\tan 33°}$

$a + b = 1$ km

$\frac{h}{\tan 57°} + \frac{h}{\tan 33°} = 1$

$h\left(\frac{1}{\tan 57°} + \frac{1}{\tan 33°}\right) = 1$

$h = \frac{1}{\left(\frac{1}{\tan 57°} + \frac{1}{\tan 33°}\right)} = 0.45677\ldots$

$\therefore$ 457 metres